IAE Nantes
Économie & Management
Pôle Sociétés

Nantes Université

Master 2 – Econometrics and Statistics
Applied Econometrics Track

Advanced Financial Econometrics

Conditional Volatility Modeling and Risk Measurement (VaR): Application to the SSE Composite Index

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Academic Year 2025–2026

Source code: github.com/floriancrochet/master-year2-advanced-financial-econometrics

Abstract

This study proposes an advanced econometric modeling of the conditional volatility of the Shanghai Stock Exchange Composite Index (SSE), with the ultimate aim of optimizing stock market risk prediction. Through the empirical estimation of eight GARCH-type and RiskMetrics specifications, under both normal and leptokurtic distributional assumptions, the study demonstrates the in-sample statistical superiority of the asymmetric GJR-GARCH(1,1) model under the Student-t distribution. This autoregressive process effectively captures both the volatility persistence and the leverage effect inherent in the Chinese stock market. Over the out-of-sample period (2025), although it reveals an adjustment inertia in the face of extreme exogenous shocks (tariff-induced crash), this specification enhances the reliability of Value-at-Risk (VaR) estimation. The model thus ensures an efficient capital allocation (Expected Shortfall) against endogenous market movements, as validated by the unconditional coverage test (Kupiec POF).

Keywords: Conditional volatility, GJR-GARCH, Value-at-Risk (VaR), Expected Shortfall (ES), SSE Index.

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Glossary of Acronyms

Acronym	Definition
ACF	Autocorrelation Function
AIC	Akaike Information Criterion
ARCH	Autoregressive Conditional Heteroskedasticity
DM	Diebold-Mariano (Test)
ES	Expected Shortfall
EWMA	Exponentially Weighted Moving Average
GARCH	Generalized Autoregressive Conditional Heteroskedasticity
GJR	Glosten-Jagannathan-Runkle (Model)
HQ	Hannan-Quinn (Information Criterion)
IGARCH	Integrated GARCH
LM	Lagrange Multiplier (Test)
MCS	Model Confidence Set
MSE	Mean Squared Error
OHLC	Open, High, Low, Close
OOS	Out-of-Sample
PACF	Partial Autocorrelation Function
PBOC	People's Bank of China
POF	Proportion of Failures (Kupiec Test)
sGARCH	Standard GARCH
SOE	State-Owned Enterprise
SSE	Shanghai Stock Exchange
STAR	Sci-Tech Innovation Board
VaR	Value-at-Risk

Introduction

This study analyzes the return dynamics of the Shanghai Stock Exchange Composite Index (SSE). Its objective is twofold: on the one hand, to model the conditional volatility of the asset over the in-sample period (2021–2024), and on the other hand, to exploit the optimal econometric specification to forecast market risk (Value-at-Risk and Expected Shortfall) over an out-of-sample horizon (2025).

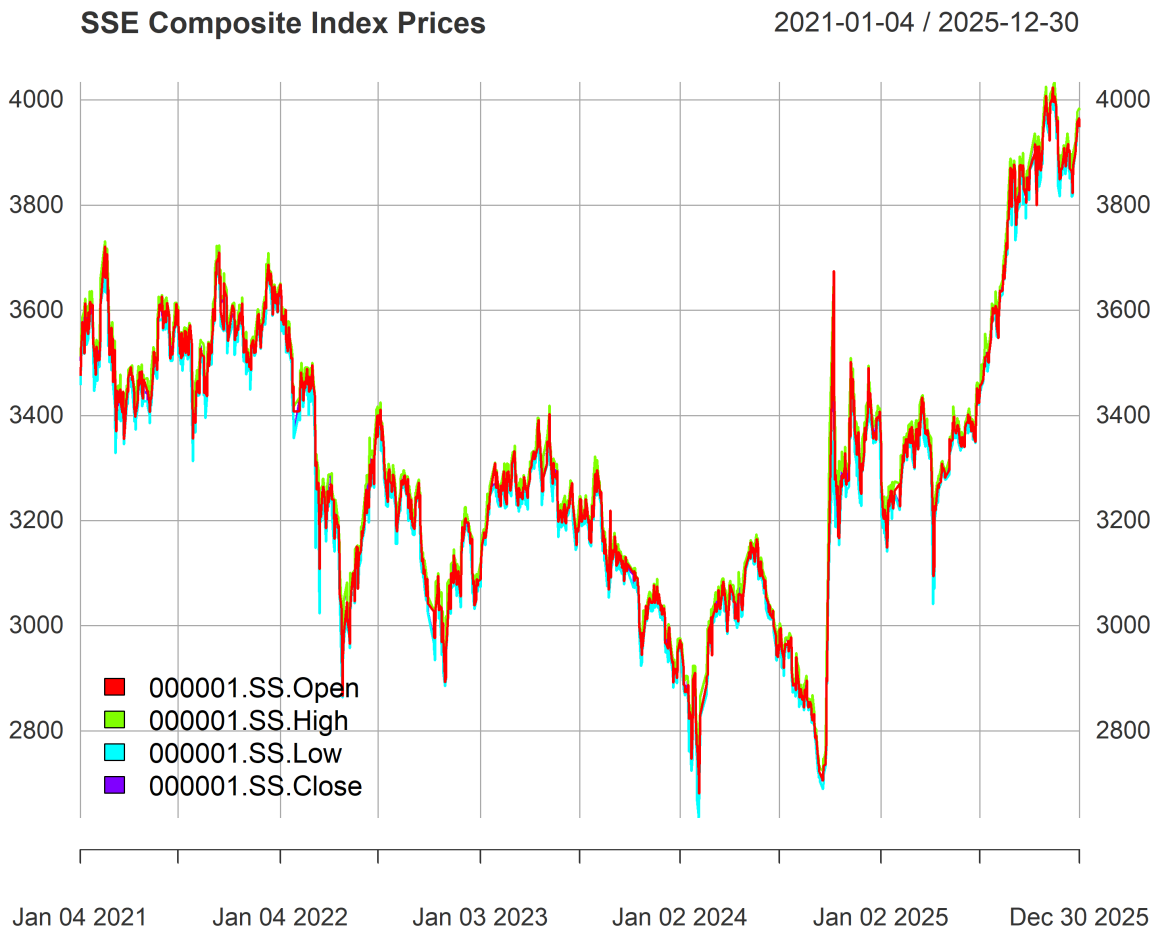
The empirical analysis relies on the historical time series of the index's daily returns. In order to capture all the stylized facts inherent in the Chinese stock market, such as volatility clustering, the leverage effect, and heavy-tailed distributions (leptokurtosis), our methodology compares different families of GARCH processes under the assumptions of Normal and Student-t distributions.

1 Exploratory and Descriptive Analysis (2021-2025)

Raw Series

The empirical analysis relies on the time series of closing prices of the SSE Composite Index (ticker `000001.SS`), observed on a daily basis over the period from January 1, 2021 to December 31, 2025. This benchmark index, weighted by total market capitalization, aggregates the performance of all type A and B shares listed on the Shanghai Stock Exchange, the leading financial marketplace in mainland China and the third-largest stock exchange globally by market capitalization. The asset is characterized by a strong predominance of the financial (27.4%) and industrial (18.7%) sectors, reflecting the structure of the Chinese state-driven economy through the massive presence of large state-owned enterprises (SOE). At the end of the 2024 fiscal year, the constituent companies of the Main Board reported a cumulative revenue of approximately RMB 49,570 billion (approx. USD 6,900 billion), attesting to the economic depth of the index. Quoted in Renminbi (CNY), the SSE Composite also incorporates the dynamism of the STAR market dedicated to technological innovation, thereby offering comprehensive exposure to the economic cycles of the world's second-largest economy. The daily historical evolution of its closing prices is illustrated in the figure below.

Figure 1: Daily evolution (OHLC) of the SSE Composite Index (2021-2025)

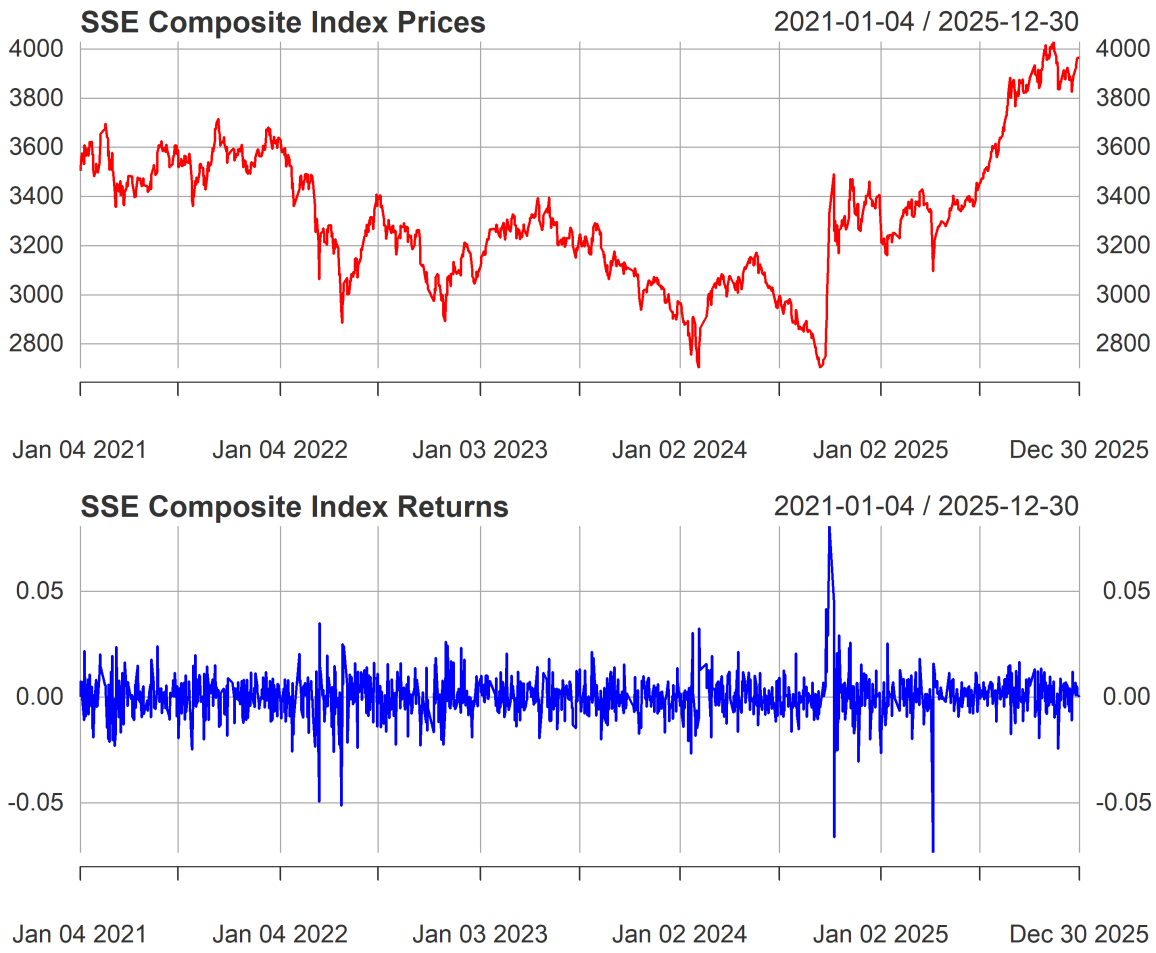


Visual inspection of the SSE Composite Index evolution between 2021 and 2025 reveals complex and successive market dynamics. The first phase, spanning from early 2021 to the beginning of 2024, is marked by a structurally bearish trend. The index depreciates from levels around 3,700 points to reach a low of approximately 2,600 points in early 2024, reflecting a prolonged contraction in investor confidence and a decelerating macroeconomic environment.

During the first three quarters of 2024, the market experiences significant instability, characterized by an intermediate rebound toward 3,150 points followed by an abrupt decline back to its annual lows (approximately 2,700 points). A phenomenal speculative breakout then occurs in late September 2024. Driven by massive state-sponsored liquidity injections, the index undergoes a sharp, straight-line appreciation, quickly surpassing 3,600 points before suffering a violent bearish correction (historic profit-taking in October) that confines the price within a broad consolidation channel between 3,200 and 3,500 points for the remainder of the year.

Finally, the year 2025 opens with continued caution before being brutally disrupted in the spring by a violent tariff-induced crash, pushing the index back down to a deep trough around 3,050 points. However, far from signaling capitulation, this sudden exogenous shock marks the starting point of a powerful structural bullish rally. The index then surges almost in a straight line, rebounding vigorously to break through and establish new valuation records above 4,000 points, thereby closing the period at its absolute periodic highs and illustrating spectacular buying pressure. In order to visualize the volatility inherent in these market movements, it is essential to compare the price evolution with the series of daily returns.

Figure 2: Evolution of closing prices and daily returns of the SSE Composite Index (2021-2025)



The joint examination of closing prices and daily returns refines our previous analysis by highlighting the pronounced volatility clustering phenomenon, a hallmark of financial time series. During the prolonged bearish phase stretching from 2021 to early 2024, the amplitude of returns remains broadly moderate and predominantly falls within a narrow corridor ranging from -2.5% to $+2.5\%$, with the notable exception of two sharp

bearish drops reaching and slightly breaching the -5% threshold in early 2022.

The abrupt regime change in the market during autumn 2024 is graphically reflected by a genuine explosion of variance. The speculative breakout observed in prices simultaneously generates, on the lower subplot, fluctuations of unprecedented magnitude: the index records in quick succession its strongest daily rebound (shattering the $+5\%$ technical threshold) and a bearish correction of rare violence, plunging below the -5% mark. Finally, the instability of the 2025 rally is confirmed on the return time series before gradually subsiding. The spring tariff-induced crash stands out conspicuously as the most extreme negative shock of the entire study period, with the daily loss plunging beyond -5% before the nervousness eventually abates during the final bullish surge.

Outlier Detection

The graphical overlay of the raw series and the series filtered of its outliers enables visualization of the magnitude of these major market shocks.

Figure 3: Raw and adjusted returns of the SSE Composite Index

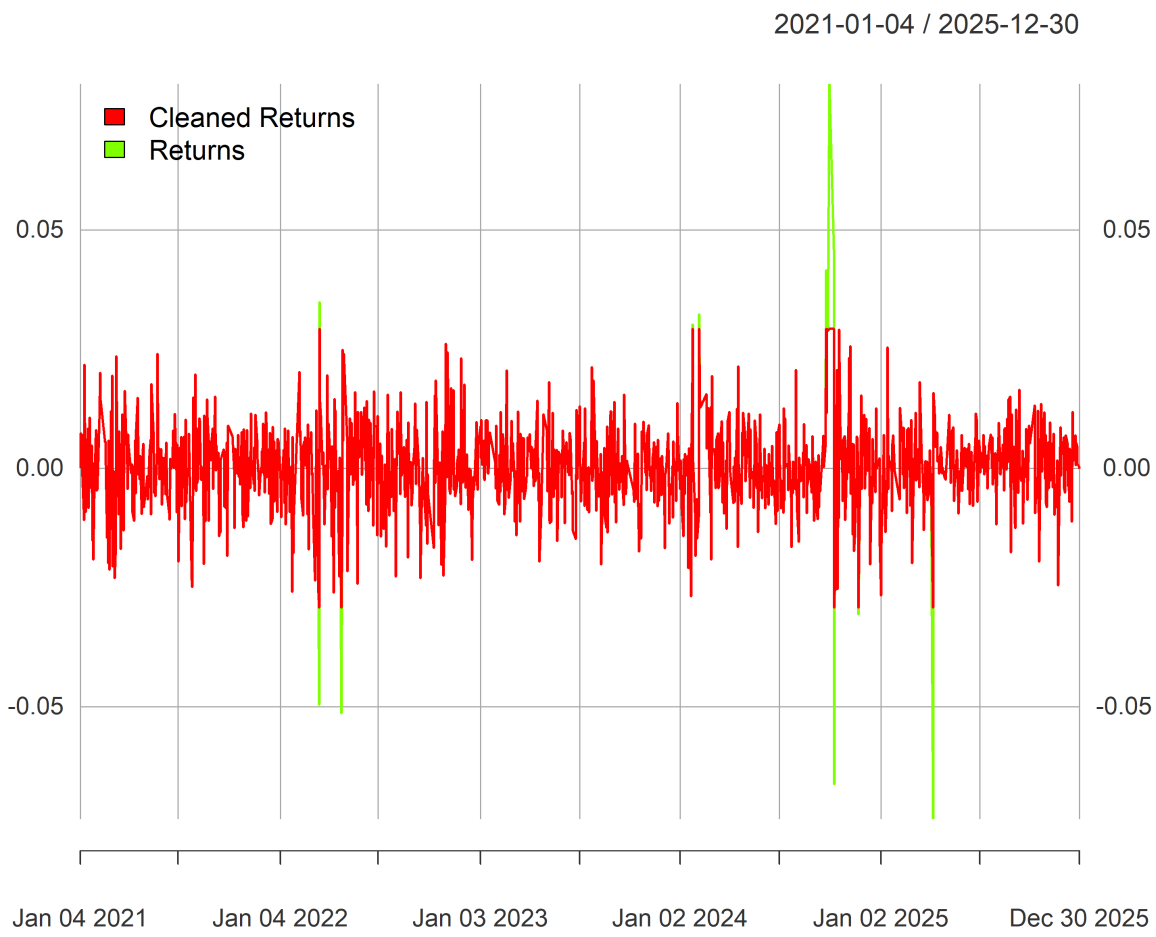


Table 1: Top 3 outliers detected in the returns of the SSE Composite Index

Date	2024-09-30	2024-10-09	2025-04-07
Correction	-0.0514	0.0370	0.0443

The analysis of these extreme values highlights three major shocks in the Chinese stock market, closely linked to the macroeconomic and geopolitical context:

The outlier of **September 30, 2024** (correction of -0.0514) reflects a historic surge in the SSE Index on the eve of the Golden Week. This positive shock is explained by

the announcement of a large-scale monetary and fiscal stimulus package by the Chinese authorities, notably including interest rate cuts by the central bank (PBOC) aimed at boosting the economy and supporting the real estate sector.

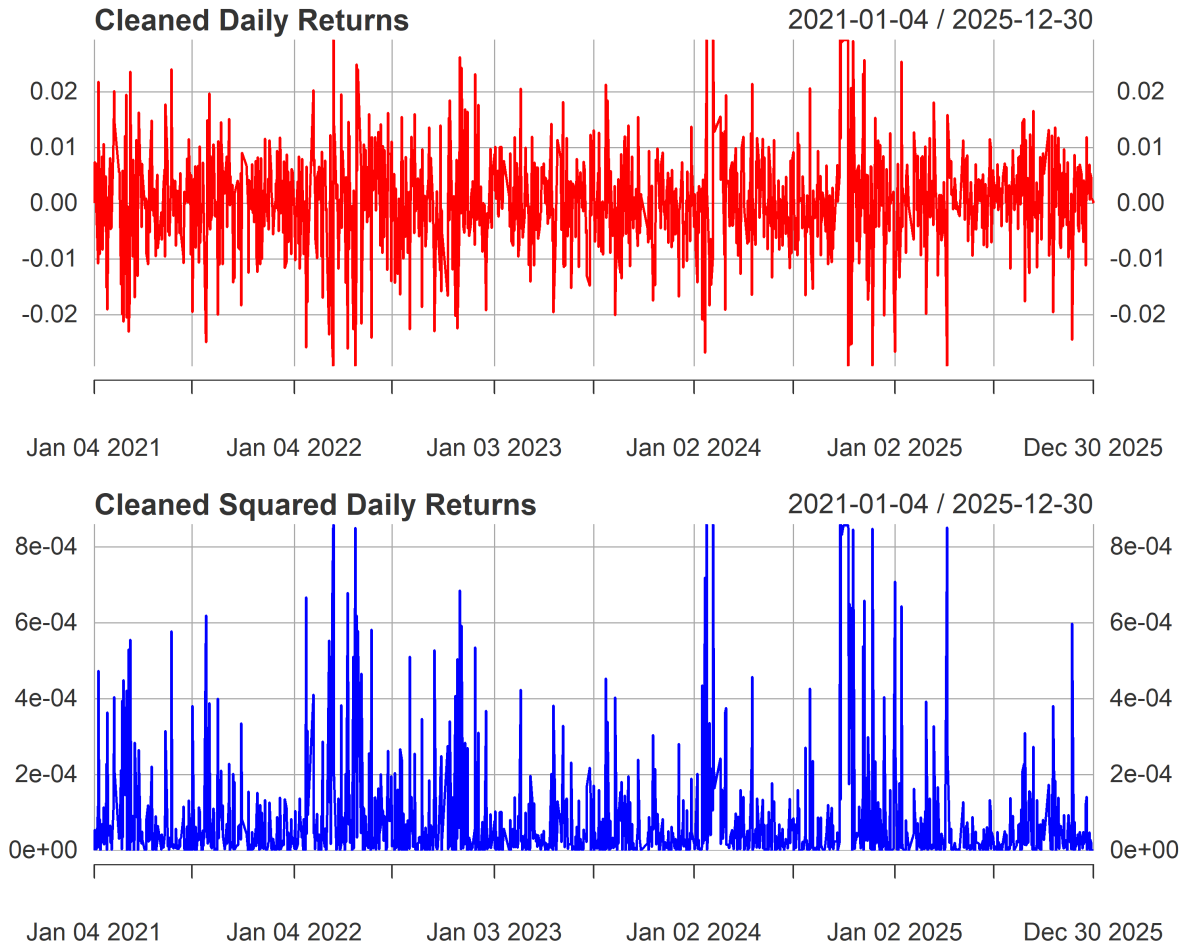
On **October 9, 2024** (correction of 0.0370), a sharp negative shock is conversely observed. Upon the reopening of markets following the national holiday, the absence of new immediate fiscal stimulus announcements dampened investor optimism. This disappointment translated into heavy profit-taking and an abrupt decline in the index.

Finally, the outlier of **April 7, 2025** (correction of 0.0443) corresponds to a severe bearish correction following the Qingming Festival. This sudden downturn was triggered by a re-escalation of Sino-American trade tensions, following the announcement of massive new tariff increases by the United States. This decision provoked a violent wave of selling across all Chinese markets. In order to avoid biasing the conditional variance modeling, the extreme impact of these three atypical events must be physically neutralized from the time series.

Adjusted Returns

The elimination of outliers stabilizes returns and allows for the visual isolation of the true dynamics of their volatility.

Figure 4: Evolution of adjusted daily returns and their squares for the SSE Composite Index (2021-2025)



Following the detection and treatment of the outliers identified in the preceding analysis, the upper subplot illustrates the dynamics of the adjusted daily return series. This econometric filtering has trimmed the amplitude of extreme values, with the series now evolving within a narrower interval whose maximum fluctuations no longer exceed the $\pm 3\%$ threshold. Although the magnitude of these events has been neutralized, the temporal dependence dynamics of the financial index are broadly preserved, ensuring the robustness of forthcoming estimations.

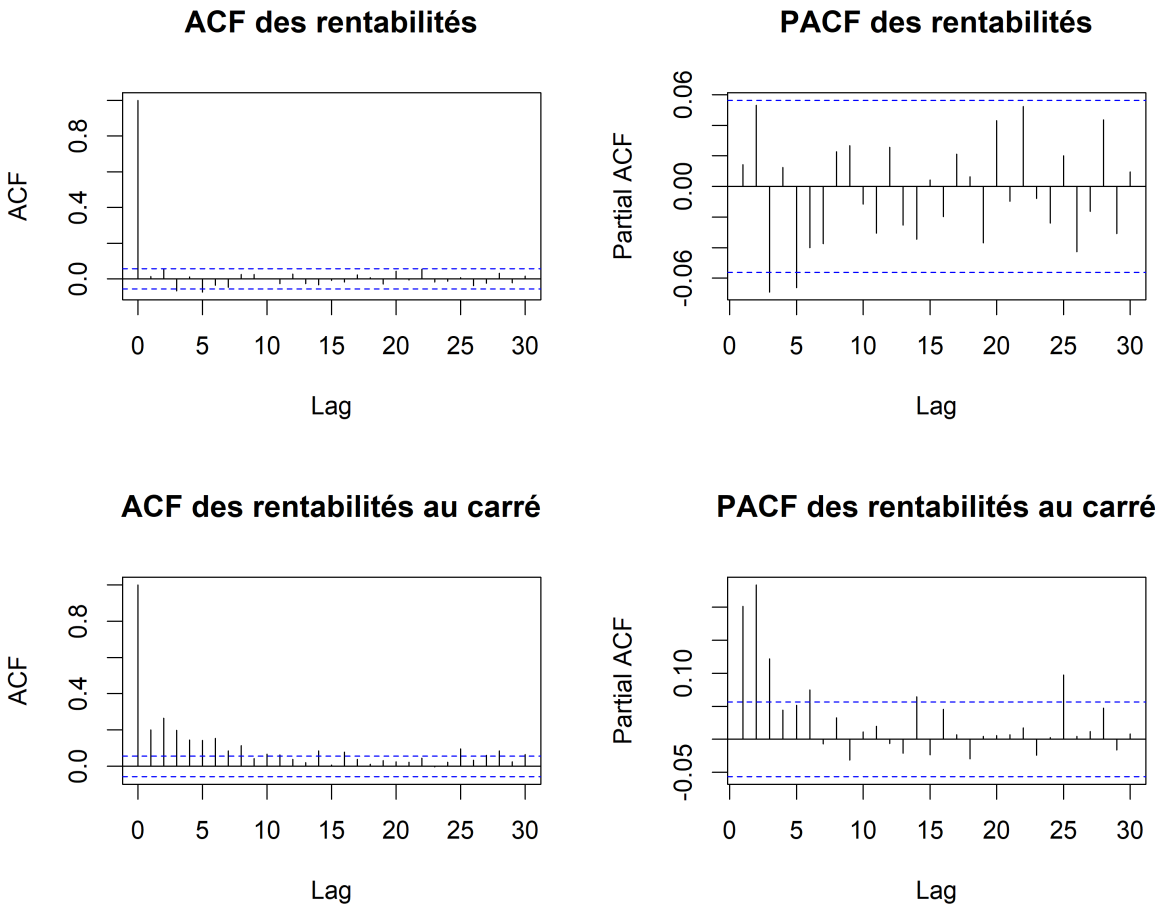
The study of this adjusted series is complemented, on the lower subplot, by the evolution of the squared adjusted return time series, which constitutes a natural empirical

proxy for the conditional variance. This plot confirms the persistence of strong conditional heteroskedasticity within the series: the major turbulence clusters previously identified (such as the September/October 2024 breakout and the April 2025 crash) materialize as clusters of variance spikes. The presence of such volatility clusters (illustrating the volatility clustering phenomenon) justifies the use of GARCH-family models envisaged for capturing the risk dynamics of the stock index. Before formally proceeding to the estimation of GARCH-type processes, this empirical suspicion of volatility clustering must be consolidated through the analysis of linear autocorrelation.

Autocorrelation

Visual inspection of the correlograms allows for the examination of linear autocorrelation in returns and the presence of ARCH effects.

Figure 5: Correlograms (ACF and PACF) of adjusted returns and their squares



Visual inspection of the correlograms highlights the temporal dependencies within the series. The autocorrelation function (ACF) and partial autocorrelation function (PACF) of the adjusted returns reveal a near-absence of autocorrelation. With the vast majority of coefficients falling within the asymptotic confidence intervals (represented by the horizontal blue dashed lines), the series graphically resembles a white noise process.

In contrast, the correlograms of squared returns reveal distinct patterns between the ACF and the PACF. The ACF displays a slow and progressive decay with strictly positive values across multiple lags, illustrating the temporal persistence of volatility. Concurrently, the PACF exhibits an abrupt drop-off and predominantly evolves within the confidence bands beyond the first few lags. This duality (ACF persistence and PACF

drop-off) constitutes the graphical signature of an autoregressive process on the squared returns, indicating a heteroskedastic conditional variance. This observation underscores that while the daily shocks appear to be uncorrelated in mean, they are by no means independent. The visual presence of these ARCH effects motivates the estimation of the heteroskedastic component through GARCH-family models, an intuition that must now be formally validated by rigorous statistical tests.

Descriptive Statistics and Tests

The formal characterization of the empirical distribution and the rigorous validation of stylized facts rely on a set of unconditional statistical tests.

Table 2: Descriptive statistics of the SSE Composite Index returns (2021-2025)

Obs.	**Mean (%)**	**Median (%)**	**Min (%)**	**Max (%)**	**Std. Dev. (%)**
1211	0.0175	0.0182	-2.9180	2.9321	0.9002
Skewness	$\boldsymbol{\nu_1}$	**Ex. Kurtosis**	$\boldsymbol{\nu_2}$	**Jarque-Bera**	**p-value**
-0.0710	-1.0087	1.1744	8.3422	70.611	0.000
Q(10)	**p-value**	**LM ARCH(5)**	**p-value**	**LM ARCH(10)**	**p-value**
21.714	0.0166	133.26	0.000	143.74	0.000

Table 2 presents the descriptive statistics of the daily returns of the SSE Composite Index. The mean return stands at 0.0175%, a value close to zero, consistent with the classical stylized facts of financial return series.

Regarding asymmetry, we examine the significance of the Skewness coefficient (-0.0710) using its test statistic $\nu_1 = -1.0087$. In absolute value, this statistic is below the critical value of 1.96 (95% confidence level). We therefore fail to reject the null hypothesis of symmetry. Statistically, the return distribution does not exhibit significant asymmetry.

The examination of tail thickness reveals a test statistic associated with the Excess Kurtosis of $\nu_2 = 8.3422$. Since this value exceeds 1.96, the coefficient of 1.1744 is statistically significant. We thus reject the null hypothesis of a mesokurtic distribution. The distribution is consequently leptokurtic (fat-tailed), implying a higher probability of extreme events than under a normal distribution.

The Jarque-Bera test corroborates this joint analysis. With a test statistic of 70.611 and a p-value of 0.000, the test overwhelmingly rejects the normality hypothesis, even at the 1% significance level. This conclusion is consistent with the significance of the Kurtosis observed previously.

Regarding linear temporal dependence, we report the Portmanteau Q(10) statistic from the Ljung-Box test applied to the return series. The test statistic amounts to 21.714, associated with a p-value of 0.0166. Although this result leads to the rejection of the null hypothesis at the conventional 5% level, it is essential to emphasize that the absence of autocorrelation is not rejected at the stricter 1% level ($0.0166 > 0.01$). This linear interdependence is therefore structurally marginal. Both statistically and economically, it does not contradict the empirical white noise diagnosis established earlier during the

visual analysis of the ACF, where the vast majority of coefficients fluctuated within the confidence bands.

Finally, the LM ARCH test at lag 10 allows testing whether the heteroskedasticity of the variance is conditional. The test statistic of 143.74, associated with a p-value of 0.000, leads to a firm rejection of the null hypothesis at the 1% level. This statistical significance indicates the confirmed presence of ARCH effects (volatility clustering), a necessary and validating condition for the estimation of GARCH-family models. Lastly, the comparison of the empirical density with a theoretical normal distribution provides an ultimate visual check of the heavy-tailedness of the distribution (leptokurtosis).

Figure 6: Empirical distribution and theoretical normal density of the adjusted returns of the SSE Composite Index

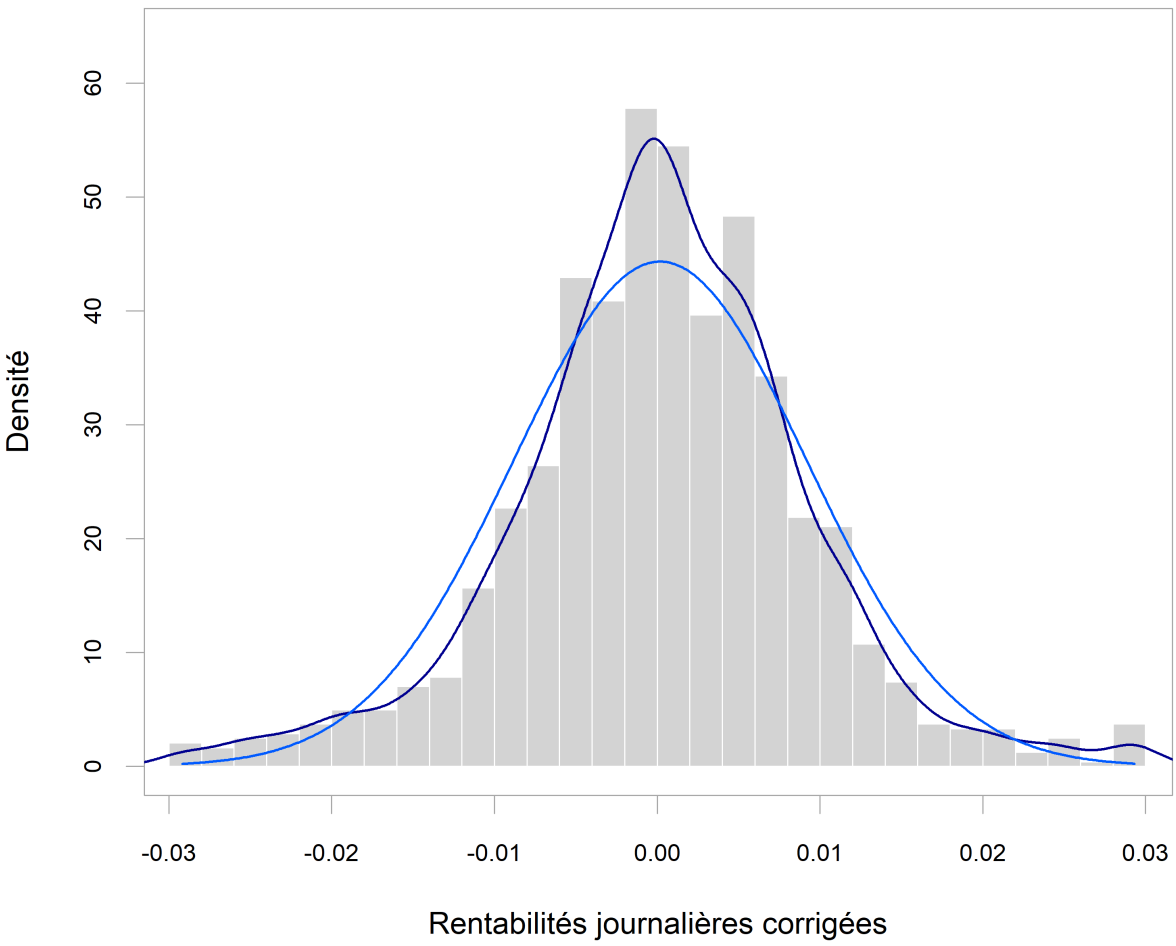


Figure 6 provides immediate visual confirmation of the statistical diagnostics established in the preceding table. The overlay of the empirical return density (dark blue curve) with the corresponding theoretical normal distribution (light blue curve) conspicuously illustrates the non-Gaussian nature of the series. The central peak is

markedly more pronounced, reflecting a strong concentration of observations around the mean, while the tails of the distribution reveal higher probabilities of occurrence than under the normal distribution.

This graphical representation materializes the mathematical concept of leptokurtosis and corroborates the overwhelming rejection of the normality hypothesis confirmed by the Jarque-Bera test. From the standpoint of the forthcoming econometric modeling, this empirical finding supports the relevance of comparing GARCH estimations under a classical normal distribution with those adopting heavy-tailed distributions (such as the Student-t distribution), which are inherently better suited to capturing extreme values.

2 Volatility Model Estimation (2021-2024)

We estimate volatility over the 2021-2024 calibration sub-period by comparing eight distinct model configurations. Our methodology relies on four specifications of the conditional variance: the GARCH(1,1), IGARCH(1,1), RiskMetrics (corresponding to an IGARCH process whose parameters are fixed a priori), and GJR-GARCH(1,1) models. In order to best capture the empirically observed leptokurtosis of returns, each of these dynamics is estimated under two distributional assumptions: the standard normal distribution and the Student-t distribution.

Figure 7: Estimation results, validation tests, and selection criteria for GARCH-family models under the normality assumption

	Coefficient		|t-value| > 1.64	persistence	half-life	log-likelihood	Akaike	HQ
GARCH	omega > 0	0.000004	4.46	0.9503	13.61	4036.93	-6.661	-6.654
	alpha ≥ 0	0.069376	6.27					
	beta ≥ 0	0.880964	88.22					
	alpha + beta < 1	0.950340						
GJR-GARCH	omega > 0	0.000004	5.75	0.9447	12.18	4040.41	-6.665	-6.657
	alpha	0.047476	2.59					
	beta ≥ 0	0.867904	77.04					
	gamma	0.058611	1.67					
	alpha + gamma ≥ 0	0.106087						
	alpha + beta + (gamma/2) < 1	0.944686						
IGARCH	omega > 0	0.000001	0.06			4028.52	-6.648	-6.644
	alpha ≥ 0	0.077967	0.38					
	beta	0.922033						
Riskmetrics	alpha	0.06				4021.94	-6.641	-6.639
	beta	0.94						

Figure 8: Estimation results, validation tests, and selection criteria for GARCH-family models under the Student-t distribution

	Coefficient		|t-value| > 1.64	persistence	half-life	log-likelihood	Akaike	HQ
GARCH	Cst(V) > 0	0.000240	1.12	0.9539	14.69	4050.53	-6.681	-6.673
	alpha ≥ 0	0.077124	2.34					
	beta ≥ 0	0.876781	35.85					
	alpha + beta < 1	0.953905						
	Student	7.907857	3.79					
GJR-GARCH	Cst(V) > 0	0.000136	0.60	0.9451	12.27	4053.32	-6.684	-6.675
	alpha	0.051080	2.19					
	beta ≥ 0	0.860186	45.33					
	gamma	0.067655	1.69					
	alpha + gamma ≥ 0	0.118735						
	alpha + beta + (gamma/2) < 1	0.945094						
	Student	8.044214	4.55					
IGARCH	Cst(V) > 0	0.000254	1.10			4045.80	-6.675	-6.669
	alpha ≥ 0	0.087596	1.33					
	beta	0.912404						
	Student	6.521546	3.19					
Riskmetrics	alpha	0.06				4040.40	-6.670	-6.666
	beta	0.94						
	Student	8.222769	5.33					

The identification of valid models rigorously rests on three required econometric criteria: compliance with the positivity constraints of the conditional variance, verification of process stationarity (characterized by a persistence strictly less than 1), and statistical significance of the estimated structural parameters (assessed by a robust statistic such that $|t| > 1.64$, corresponding to a minimum significance level of 10%). It should be noted that the potential non-significance of the constant ω in the variance equation does not, by itself, constitute grounds for model invalidation, insofar as this parameter serves as an anchor for the unconditional variance level and does not affect the conditional dynamics per se.

Against these criteria, the GARCH(1,1) and GJR-GARCH(1,1) models estimated under the normal distribution assumption unequivocally satisfy all conditions. Indeed, their coefficients guarantee a strictly positive variance ($\omega > 0$, $\alpha \geq 0$, $\beta \geq 0$, and $\alpha + \gamma \geq 0$) as well as a stationary process (persistence of 0.9503 for GARCH and 0.9447 for GJR-GARCH). Moreover, all of their empirical parameters prove to be statistically significant.

The adoption of a Student-t distribution enriches the modeling by explicitly capturing the leptokurtosis of returns, as evidenced by the strong significance of the shape parameter in each specification. Under this distributional assumption, the GARCH(1,1)-t and GJR-GARCH(1,1)-t models remain valid: their structural parameters α, β, and γ are all significant, the positivity constraints are satisfied, and the processes remain stationary (persistence of 0.9539 for GARCH-t and 0.9451 for GJR-GARCH-t). Only the constant ω loses significance ($t = 1{,}12$ and $t = 0{,}60$ respectively), which, in accordance with the criterion stated above, does not call into question the validity of these specifications.

In contrast, both IGARCH(1,1) models, whether estimated under the normal or Student-t distribution, are invalidated due to the non-significance of their ARCH component α ($t = 0{,}38$ under the normal distribution and $t = 1{,}33$ under the Student-t distribution). This structural parameter, which governs the response of conditional variance to past shocks, constitutes a fundamental element of the model dynamics, and its statistical insignificance compromises the econometric interpretation of the IGARCH specification.

Finally, the RiskMetrics specifications, whose parameters are imposed exogenously (with $\alpha = 0.06$ and $\beta = 0.94$), are by construction not subject to significance tests. These specifications model an integrated process (unit persistence) and inherently violate the strict stationarity assumption. Nevertheless, as widely used industry-standard regulatory benchmarks, they are retained in the analysis as benchmark references, while bearing in mind this theoretical limitation.

In total, six models are retained for the remainder of the analysis: GARCH(1,1) and

GJR-GARCH(1,1) under the normal distribution, GARCH(1,1)-t and GJR-GARCH(1,1)-t under the Student-t distribution, as well as both RiskMetrics specifications (normal and Student-t).

Table 3: Residual diagnostics for the GARCH-type models

Model	**Q(5)**	p**-value**	$\boldsymbol{Q^2}$**(5)**	p**-value**	**LM-ARCH(5)**	p**-value**
GARCH (Normal)	2.89	0.427	3.26	0.362	1.33	0.637
GJR-GARCH (Normal)	3.24	0.364	3.76	0.285	1.57	0.575
RiskMetrics (Normal)	2.79	0.447	7.35	0.043	4.44	0.139
GARCH (Student)	2.76	0.453	2.20	0.573	1.25	0.659
GJR-GARCH (Student)	3.15	0.380	2.77	0.450	1.31	0.645
RiskMetrics (Student)	2.80	0.445	7.27	0.045	4.35	0.145

Table 3 presents the results of diagnostic tests applied to the standardized residuals, determining the overall post-estimation validity of the models. A volatility model is deemed statistically adequate if it succeeds in capturing all of the temporal dynamics of the series, yielding standardized residuals that formally behave as white noise.

First, the examination of the Ljung-Box Portmanteau $Q(5)$ statistic, which tests for residual linear autocorrelation, reveals p-values uniformly and clearly above the conventional 5% significance level (ranging from 0.364 to 0.453). The null hypothesis of no autocorrelation is not rejected for any of the six specifications at the mean level.

Second, the effectiveness of the variance modeling itself is assessed through the Ljung-Box test on squared standardized residuals $Q^2(5)$ and the Lagrange Multiplier LM-ARCH(5) test. For the freely estimated models, namely the GARCH(1,1) and GJR-GARCH(1,1) (whether based on a normal or Student-t distribution), all p-values from these tests convincingly exceed the critical threshold of 0.05 (ranging from 0.285 to 0.659). We thus validate the null hypothesis confirming the absence of residual ARCH effects. From an econometric standpoint, their conditional variance equations are perfectly specified since no residual volatility clustering persists among their errors.

However, the diagnosis is far more critical for the two RiskMetrics benchmark models. Their $Q^2(5)$ tests reveal the lingering presence of dependence effects in the squared residuals (p-values of 0.043 and 0.045 respectively, indicating a rejection of the variance independence hypothesis at the 5% level). This shortcoming highlights the fundamental weakness inherent in the EWMA models used by RiskMetrics: the arbitrary and universal imposition of fixed smoothing parameters ($\alpha = 0.06$ and $\beta = 0.94$). By failing to calibrate its coefficients on local historical data via maximum likelihood, the

prescriptive RiskMetrics approach fails to fully filter the heteroskedasticity specific to the asymmetric endogenous shocks of the Chinese market (SSE).

In conclusion, only the GARCH(1,1) and GJR-GARCH(1,1) models, under both distributional specifications (normal and Student-t), whose structural parameters have been fully estimated, demonstrate rigorous adequacy with local volatility, enabling their use for out-of-sample forecasting.

Among all of these statistically justified specifications (across both distributions), the selection of the optimal model logically proceeds on the basis of information criteria and the likelihood function. In this regard, the GJR-GARCH(1,1) model under the Student-t distribution unquestionably emerges as the best-performing configuration. Indeed, it achieves the highest log-likelihood while strictly minimizing the AIC (Akaike) and Hannan-Quinn (HQ) criteria relative to all competing models.

The superiority of this model lies in its dual ability to capture the complex stylized facts of the SSE Index. At the level of volatility dynamics, the GJR structure allows the internalization of the asymmetric effect of shocks: the significance of its asymmetry parameter (γ) confirms the inherent empirical presence of a leverage effect in the market, attesting that negative shocks (market declines) induce a higher variance than positive shocks of equal magnitude. This autoregressive process moreover remains solidly stationary, with a persistence estimated at 0.9451 (a very high level, yet strictly below unity). At the distributional level, the use of the Student-t distribution overcomes the limitation of the classical normal distribution: through the strong significance of its shape parameter (degrees of freedom), the model perfectly absorbs the substantial probabilities of extreme events (leptokurtosis) inherent in the past crises identified during the descriptive analysis.

Finally, the post-estimation examination underscores the purity of its standardized residuals: cleansed of any linear structure (p-value of 0.380 for $Q(5)$) and freed from any residual volatility clustering (p-values of 0.450 for $Q^2(5)$ and 0.645 for LM-ARCH), the errors of this model confirm that the chosen architecture captures the entirety of the hidden dynamics in the asset's returns. Bolstered by this modeling excellence, combining the ability to reproduce cyclical asymmetry with the robustness of the temporal distribution's tails, it is the GJR-GARCH(1,1) Student-t model that is rightfully designated for the estimation of future VaR.

3 Volatility Forecasting (2025)

Before addressing financial risk measurement, it is necessary to assess the intrinsic quality of the variance forecasts generated by our models over the target time horizon (2025). The following table compares the out-of-sample predictive performance of the four validated specifications.

Figure 9: Out-of-sample (OOS) volatility forecasting performance evaluation and model comparison tests for the validated models (2025)

	Normal		Student	
	GARCH	GJR	GARCH	GJR
MSE (*10^5)	0.0127	0.0127	0.0127	0.0127
R2OOS	-	0.000	0.000	0.000
DM p-value	-	0.7517	0.0892	0.9176
MCS_R p-value	0.9340	1.0000	0.0972	0.1518
Rank_R	2	1	4	3

The evaluation of out-of-sample forecast accuracy reveals extremely homogeneous performance across the four econometric specifications. In terms of forecast error, all models display a rigorously identical mean squared error (MSE) ($0{,}0127 \times 10^{-5}$). Similarly, the virtually zero R^2_{OOS} statistics (0.000) of the alternative models underscore that none of the asymmetric or leptokurtic architectures manages to outperform the standard GARCH benchmark model (under the normal distribution) in explaining out-of-sample variance over time.

The statistical evaluation tests of this accuracy unequivocally corroborate this strong similarity. The Diebold-Mariano (DM) test, comparing each specification against the normal GARCH model, yields p-values well above the 5% significance level (ranging from 0.0892 to 0.9176). The null hypothesis of equal predictive ability therefore cannot be rejected for any of the candidates. The Model Confidence Set (MCS) test reaches the same conclusion: by displaying p-values between 0.0972 and 1.0000, the algorithm fails

to significantly discriminate among the processes and retains all four models within its superior confidence set. Nevertheless, within this statistically indistinguishable group, the ranking (Rank_R) attributes a very slight empirical advantage to the GJR-GARCH model under the normal distribution (p-value of 1.0000 and rank 1).

Having validated the robustness and homogeneity of these volatility forecasts, we can now exploit these conditional variances to estimate our primary risk metric: the Value-at-Risk.

4 VaR Forecasting (2025)

We perform an out-of-sample forecast of the Value-at-Risk (VaR) at the 5% level for the year 2025.

Out-of-Sample Dynamics

Visual analysis constitutes the first step in conditional risk evaluation. The four graphs below overlay the daily returns realized in 2025 with the Value-at-Risk (5%) thresholds projected by our specifications. This graphical comparison allows for an empirical assessment of the responsiveness of risk bounds to volatility shocks occurring out-of-sample.

Figure 10: Out-of-sample comparison of daily returns of the SSE Composite Index and the Value-at-Risk (5% VaR) estimated by the GARCH(1,1) model (2025)

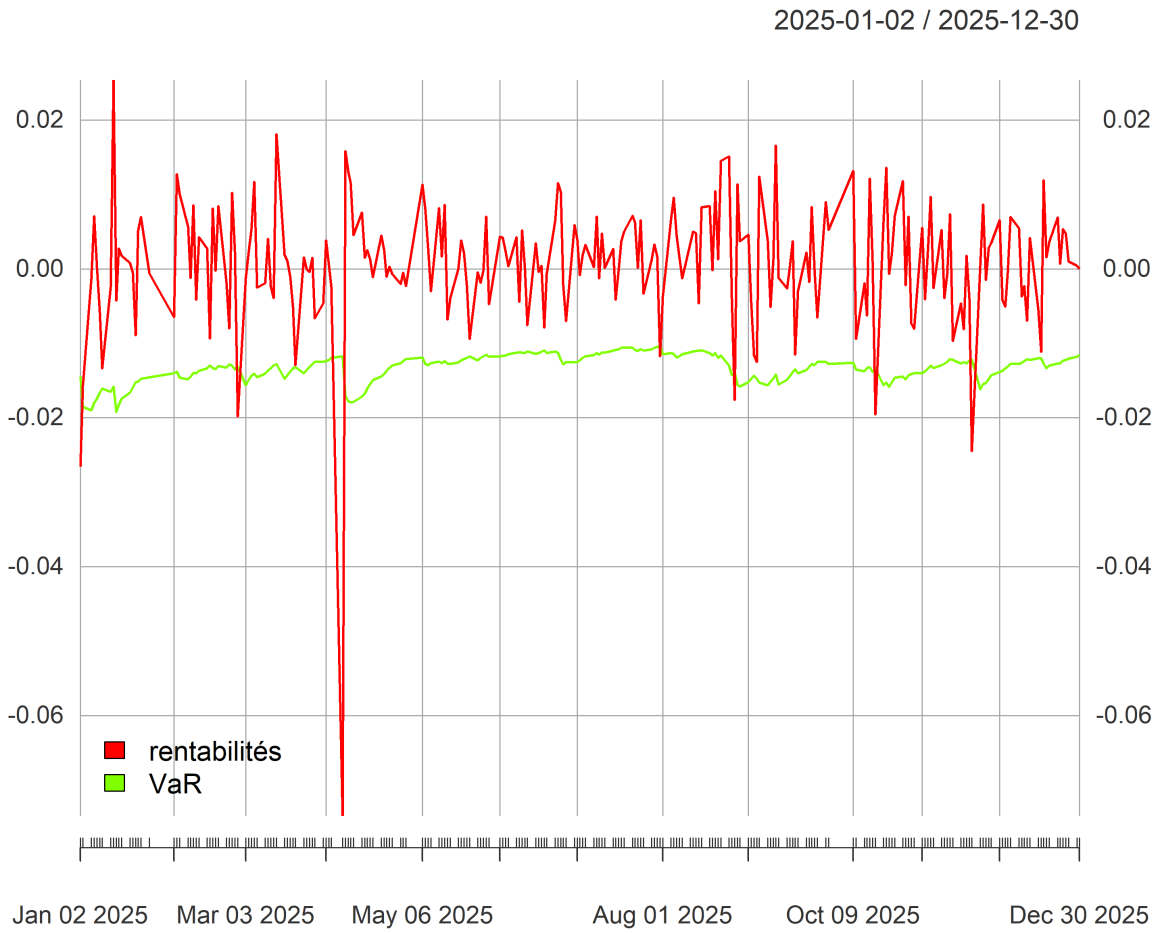


Figure 11: Out-of-sample comparison of daily returns of the SSE Composite Index and the Value-at-Risk (5% VaR) estimated by the GJR-GARCH(1,1) model (2025)

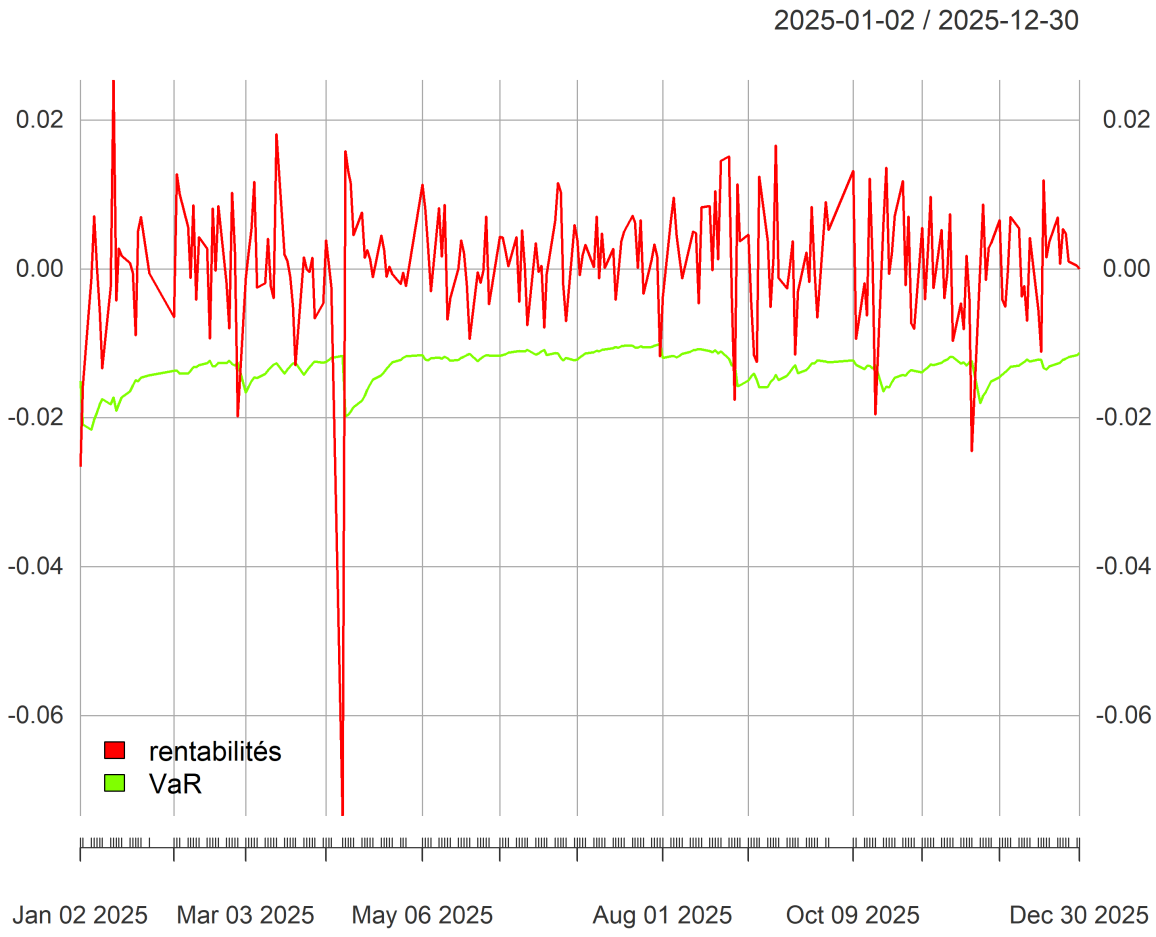


Figure 12: Out-of-sample comparison of daily returns of the SSE Composite Index and the Value-at-Risk (5% VaR) estimated by the GARCH(1,1) model under the Student-t distribution (2025)

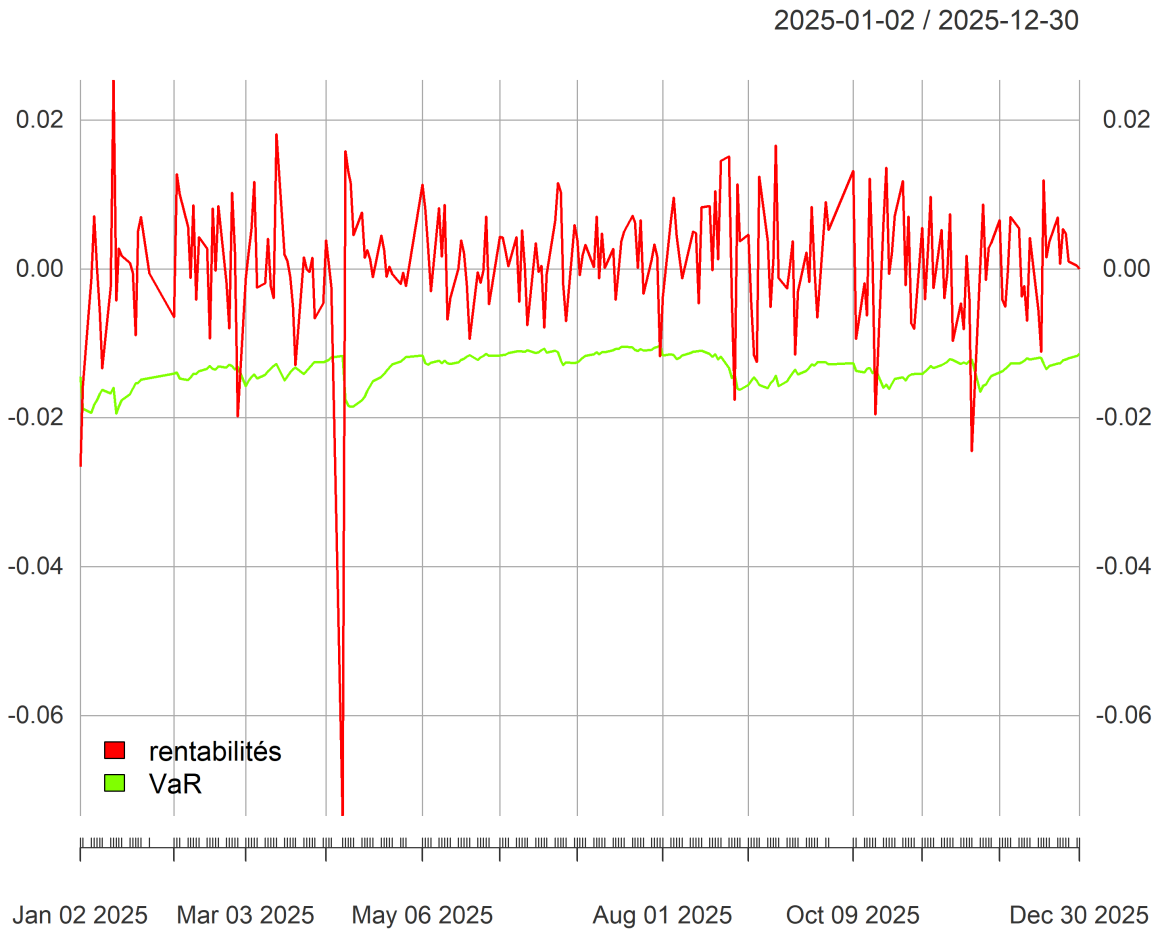
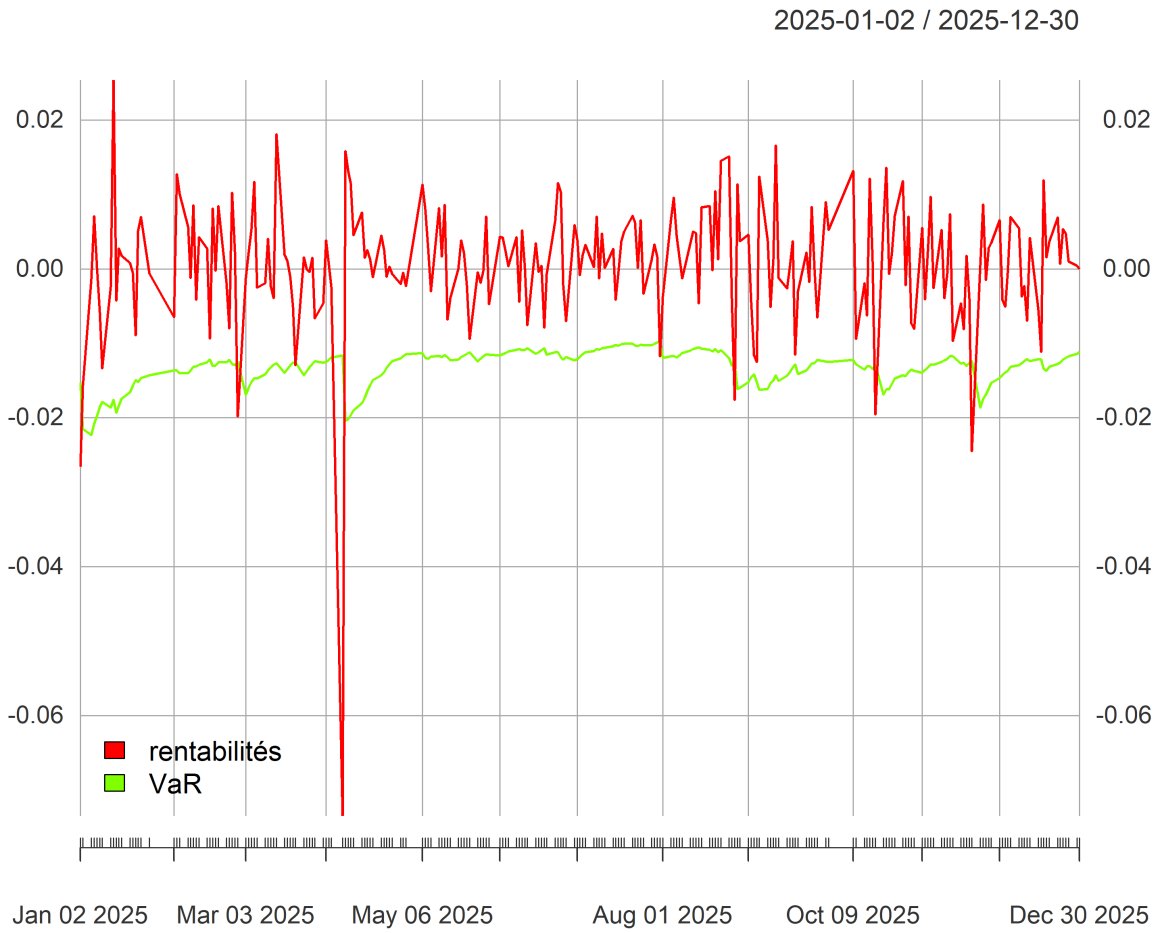


Figure 13: Out-of-sample comparison of daily returns of the SSE Composite Index and the Value-at-Risk (5% VaR) projected by the GJR-GARCH(1,1) model under the Student-t distribution (2025)



Visual inspection of the out-of-sample forecasts enables the identification of the most spectacular exception (VaR breach) of the year 2025 on the SSE Index. This extreme event occurs simultaneously across all models on April 7, 2025, when daily returns brutally collapse below the $-7\,\%$ mark, massively piercing the risk limit covered by the $5\,\%$ VaR (then gauged at approximately $-2\,\%$ by the specifications).

From an economic standpoint, as highlighted in the exploratory analysis, this abrupt decline corresponds to the crash following the Qingming Festival. This stock market downturn was driven by the sudden and exogenous announcement of a re-escalation of Sino-American trade tensions through the imposition of new tariff barriers.

From an econometric standpoint, this breakdown highlights the irreducible limitation of parametric financial modeling: a GARCH-family model, structurally conditioned by the history of past volatility, cannot anticipate the occurrence or the magnitude of

an unprecedented geopolitical information shock. Conversely, all graphs brilliantly illustrate the fundamental mechanics of conditional risk updating. Upon the realization of this extreme shock, its squared error term is integrated via the autoregressive (ARCH) component into the variance equation. However, the visual adjustment of the VaR in the aftermath of the crash remains moderate (dropping to approximately -2 % for the asymmetric GJR-GARCH models versus -1.8 % for the sGARCH models). Far from plunging spectacularly to encapsulate the shock, the VaR limit instead demonstrates a degree of inertia, underscoring the limitations of these models when confronted with volatility smoothing in the event of an extreme and unpredictable occurrence.

While this visual analysis confirms the overall responsiveness of the models, it remains insufficient to formally determine their regulatory validity. It is therefore imperative to subject these projections to a rigorous statistical validation protocol, in order to ascertain that the empirical frequency of these exceptions does not exceed the theoretical coverage threshold.

Backtesting and Capital Allocation

In order to consolidate the preceding graphical analysis and to determine the optimal capital allocation, it is necessary to formally quantify the efficiency of the models through a backtesting procedure. The table below summarizes these out-of-sample statistical evaluations, cross-referencing the exception count with the Kupiec unconditional coverage test, while financially projecting these risk measures (VaR and Expected Shortfall) onto a representative portfolio of 100,000 euros.

Figure 14: Out-of-sample backtesting (5% VaR) and risk measurement (Expected Shortfall) of the GARCH and GJR-GARCH models for the SSE Index (2025)

		Normale		Student	
		GARCH	GJR	GARCH	GJR
exceptions théoriques		12	12	12	12
exceptions empiriques		7	8	7	8
Kupiec	test	2.65	1.65	2.65	1.65
	p-value	0.104	0.199	0.104	0.199
	Mean(VaR)	-1.33%	-1.32%	-1.33%	-1.32%
	ES	-1.66%	-1.66%	-1.67%	-1.66%
Couverture portefeuille de 100 000 euros		1,660 €	1,660 €	1,670 €	1,660 €

The analysis of this backtesting table confirms the overall robustness of the risk projections. While it reveals a very high degree of homogeneity in capital allocation requirements, it nevertheless allows the asymmetric (GJR) architecture to be favored from a statistical validation standpoint (maximum p-value of 0.199), although it fails to differentiate the underlying distributional assumption (Normal or Student-t distribution), whose results converge.

First, over the 2025 horizon, the number of empirical exceptions (7 for the standard dynamics, 8 for the asymmetric GJR structures) proves to be systematically lower than the number of theoretically expected exceptions (12 for a 5% risk threshold). This cyclical

overestimation of volatility over the out-of-sample year numerically demonstrates that the models do not underestimate portfolio exposure. Statistically, this finding is formally validated by the Kupiec unconditional coverage backtest. With p-values ranging between 0.104 and 0.199, above the conventional 5% significance level, the null hypothesis of unconditional coverage accuracy cannot be rejected. All four specifications therefore offer a level of coverage that is adequate from a regulatory standpoint relative to the actual movements of the Chinese market.

The cross-sectional analysis of the metrics further underscores a very high degree of homogeneity across the estimation processes. The average VaR values converge very closely, all falling between $-1.32\,\%$ and $-1.33\,\%$. Moreover, conditional on a VaR breach, the Expected Shortfall (ES) measure also remains highly consistent at approximately $-1.66\,\%$, illustrating that the average loss incurred during an extreme trading session exceeds the simple VaR breach threshold by approximately 33 basis points. It should nonetheless be noted that this average loss severity of $-1.66\,\%$ is strongly driven upward (in absolute value) by the inclusion of the isolated extreme event of April 7, 2025, highlighting the high sensitivity of the empirical Expected Shortfall to exogenous extreme values.

Nevertheless, it is within the framework of applied portfolio management for a 100,000-euro portfolio that a model choice must be made. The risk manager's constraint is twofold: to ensure the accuracy of the unconditional coverage of the forecasts (via the Kupiec test) while minimizing the capital set aside against market shocks. Although the minimum allocation of 1,660 euros is shared by three of the four processes studied, the asymmetric GJR-GARCH family clearly stands out by guaranteeing the most robust empirical adequacy (maximum p-value of 0.199). However, backtesting reveals a perfect convergence between its two specifications (Normal and Student-t), which generate equivalent out-of-sample results at the retained level of financial precision. The final choice of a single model (such as GJR-Student-t) will therefore rest on its demonstrated statistical superiority over the in-sample estimation period, notably through the maximization of the log-likelihood and the joint minimization of information criteria (AIC, HQ).

Conclusion

This study has demonstrated that traditional parametric modeling, restricted to assumptions of normality and symmetry, proves insufficient to capture the structural complexity of the Chinese stock market in-sample. Our estimation work has formally established that the SSE Composite Index is governed both by an asymmetric heteroskedastic dynamic and by strong leptokurtosis.

While the out-of-sample forecasts for the year 2025 revealed a homogenization of predictive performance among the previously validated models, it is ultimately on the basis of its in-sample estimation statistical superiority (maximization of the likelihood, minimization of AIC and HQ criteria) that the GJR-GARCH(1,1) Student-t model has emerged as the final compromise.

The application of this model enhances the reliability of capital allocation: validated by the Kupiec test and displaying an efficient capital charge (Expected Shortfall), it optimizes risk management. However, the tariff-induced crash of April 2025 underscores the structural limitation of autoregressive smoothing (GARCH) in the face of sudden exogenous shocks: although well-suited for daily risk monitoring, these parametric models require coupling with stress tests to guard against unpredictable extreme shocks.

Appendices

Appendices (supplementary tables, figures, and R code excerpts) will be included in a forthcoming version of this report.